# Synthesis of 4,4,4-Trifluoro-1-(4-methyl-phenyl)-butane-1,3-dione through a Claisen Condensation Mechanism

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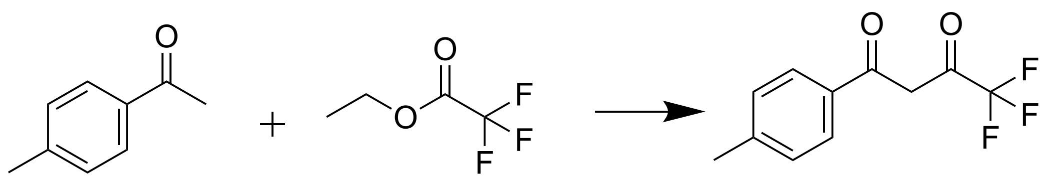

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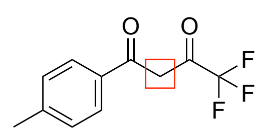
**ABSTRACT:** 4,4,4-trifluoro-1-(4-methyl-phenyl)-butane-1,3-dione is an important precursor in the synthesis of the popular non-steroidal drug Celecoxib. Traditional synthesis of this compound typically involves large quantities of toxic solvents and high amounts of heating, resulting in a not very green synthesis mechanism.[1] A novel synthesis of 4,4,4-trifluoro-1-(4-methyl-phenyl)-butane-1,3-dione was proposed by Francis Ho, James Sherwod, and Glenn A. Hurst in their paper "A Greener Synthesis of the Nonsteroidal Anti-inflammatory Drug Celecoxib" using a room temperature environment and minimal toxic reagents.[2] This synthesis mechanism is attempted to be replicated in an undergraduate laboratory setting, and modifications are proposed to improve synthesis success rates.

Celecoxib is a popular non-steroidal, anti-inflammatory drug used to treat acute pain and arthritis.[3] The patented synthesis of Celecoxib was introduced in 2008 in a patented synthesis mechanism.[1] In synthesizing the Celecoxib intermediate 4,4,4-trifluoro-1-(4-methyl-phenyl)-butane-1,3-dione **(1)**, the patent recommended using suitable inert solvents including acetone, ethyl acetate, THF, and toluene.[1] The reaction was also recommended to be carried out at any temperature range from 25°C to the solvent's reflux temperature.[1]

The initial proposed synthesis utilizes 2,2,5,5-Tetramethyloxolane (TMO) as a solvent replacement for toluene.[2] Toluene is cited by the EPA as a reproductive hazard, and possibly an air pollutant.[4] In this synthetic process, researchers performed a Claisen condensation between 4'-methylacetophenone and ethyl trifluoroacetate under room temperature conditions. Sodium methoxide was used as the base, and TMO was the solvent. After 1 hour of stirring, the product was isolated under vacuum for 40 minutes to sufficiently dry the product and remove excess solvent.[2] The product was then quantified using $H^1$-NMR, $C^{13}$-NMR, and IR spectroscopy. The most critical peak is the singlet around 6.0 ppm, which represents the two hydrogens on the carbon between the diketone in the product. The presence of this peak will be the biggest determinant of whether the product is present in the reaction **(Scheme 1)**.

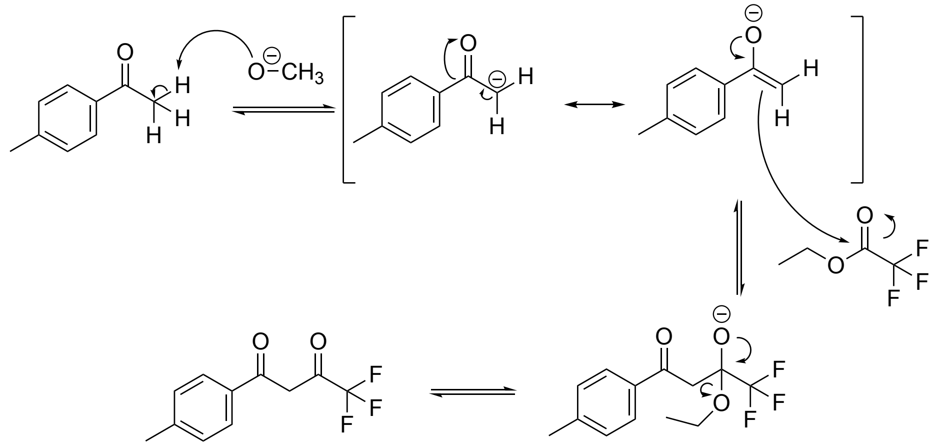
**Expected $^1$H NMR** (chloroform-d, 400MHz) δ ppm: 7.71 (d, 2H, $J = 8.2$ Hz, Ar-H), 7.19 (d, 2H, $J = 8.2$, Ar-H), 5.96 (s, 1H, $COCH_2CO$), and 2.32 (s, 3H, $CH_3$).

**Scheme 1.** Critical peaks in H-NMR characterization of desired product



The initial motivation for this synthesis examination is to successfully replicate the greener method of synthesizing **(1)** in an undergraduate laboratory setting.[5] A dry environment and acid quenching step were tested as possible synthesis mechanism modifications, and several future research points are proposed to continue to improve this synthesis procedure. To synthesize the product **(1)**, 3 separate reaction mechanisms were attempted.

**Scheme 2:** Original arrow-pushing mechanism as proposed by Francis Ho, James Sherwood, and Glenn A. Hurst.



The first reaction mechanism replicated the reaction mechanism proposed in the paper written by Ho, Sherwood, and Hurst **(Scheme 2).** Sodium methoxide (0.00344 mol), toluene (1.5mL), ethyl trifluoroacetate (0.34mL, 0.0029 mol), and 4'-methylacetophenone (0.34mL, 0.0025 mol) were combined in a 6-dram vial and then allowed to stir for 1 hour at room temperature (25°C). The product was then isolated under vacuum for 40 minutes, and any excess solvent was blown off with nitrogen.[2]

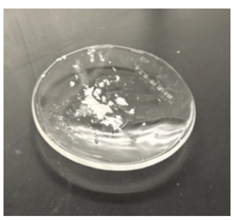


**Figure 1.** Isolated product after vacuum and nitrogen blow.

**$^1$H NMR** (400MHz, $D_2O$) δ ppm: 8.3 (s, 2H, Ar-H), 7.2 (d, 2H, *J* = 8.3, Ar-H), 6.5 (m, 7H, Ar-H), 2.5 (s, 3H, $CH_3$), 2.3 (s, 3H, $CH_3$), 2.1 (s, 3H, $CH_3$), 1.1 (t, 2H, *J* = 7.2, $CH_3$)

Upon further analysis, it was noted that a possible side reaction that could be occurring could be an acid-base reaction between sodium methoxide and water. The pKa of water is 14 and the pKa of methoxide is 16.[6,7] Since methoxide is the weaker acid, the proton transfer reaction from water to methoxide is favorable and proceeds quickly **(Scheme 2).**

The second reaction mechanism utilized the same reaction mechanism from Ho, Sherwood, and Hurst and modified it by adding a dry environment. The objective of this modification was to avoid a side reaction between sodium methoxide and water **(Scheme 3A)**. The same reaction as above was performed under a dry environment using a septum cap on a 6-dram vial. Sodium methoxide (0.00344 mol) and a stir bar were added to a 6-dram vial, and then quickly capped with a septum cap. Toluene (1.5mL), ethyl trifluoroacetate (0.34mL, 0.0029 mol), and 4'-methylacetophenone (0.34mL, 0.0025 mol) were added via a needle syringe, and the reaction was allowed to stir for 1 hour at room temperature. The product was then similarly isolated under vacuum for 40 minutes, and any excess solvent was blown off with nitrogen.[2]

During the mixing process, as seen in Figure 2, the reagents never formed a homogenous mixture, suggesting that not all reagents successfully dissolved in solution and participated in the reaction.

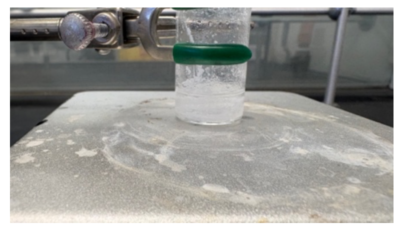


**Figure 2.** Combination of all reaction components in a 6-dram vial.

**$^1$H NMR** (400MHz, $D_2O$) δ ppm: 8.3 (s, 10H, Ar-H,$CH_3$), 7.8 (d, 2H, *J* = 8.1, Ar-H), 7.3 (d, 2H, *J* = 7.0, Ar-H), 2.6 (s, 3H, $CH_3$), 2.3 (s, 3H, $CH_3$)

Further research into the Claisen Condensation reaction notes a lack of an acid-quenching step in the original synthesis mechanism proposed by Ho, Sherwood, and Hurst.[8]

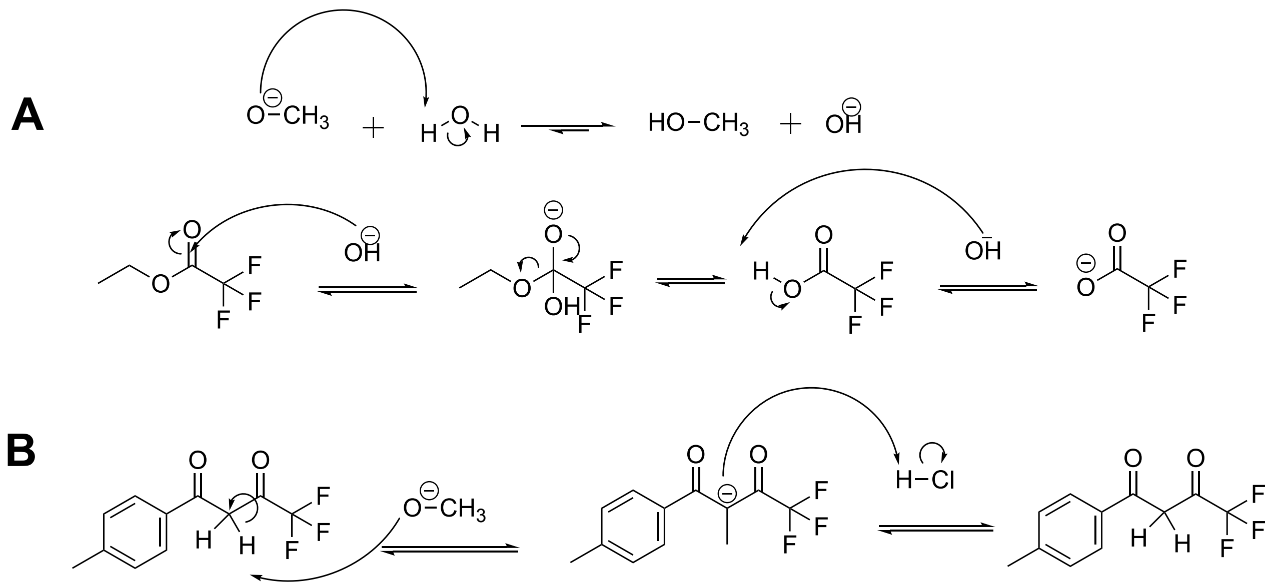
The final protonation step using HCl is irreversible, making the reaction unable to move backwards in equilibrium **(Scheme 3B).**

The third reaction mechanism referenced another synthesis mechanism proposed by Anumula, Gilla, Alla, Akki, and Bojja.[1] This mechanism was the original patent method of synthesizing Celecoxib. In this synthesis mechanism, 4'-methylacetophenone (0.85mL, 0.0064 mol), ethyl trifluoroacetate (1.19mL, 0.0099 mol), and toluene (2mL) were combined in a 50mL Erlenmeyer flask. Simultaneously, sodium methoxide (0.4027g, 0.0075) and toluene (2mL) were combined in a 6-dram vial under a dry environment, and allowed to stir for 10 minutes at 35°C. The 4'-methylacetophenone, ethyl trifluoroacetate, and toluene mixture was then added to the 6-dram vial, and then the entire reaction was allowed to stir at 75°C for 4 hours. Once the reaction finished the stirring period, it was allowed to sit overnight at room temperature. The resulting mixture was then quenched with 2mL of water and 3mL of 20% HCl. The aqueous layer of the product was washed with 2x2mL of toluene, and then the organic solvent was removed by rotary evaporator at 55°C.[1] The isolated product had two components: a translucent, slightly yellow viscous liquid, and a solid, waxy white product.

**Liquid product: $^1$H NMR** (400MHz, $CDCl_3$) δ ppm: 7.9 (d, 2H, *J* = 8.2, Ar-H), 7.3 (d, 2H, *J* = 8.0, Ar-H), 2.6 (s, 3H, $CH_3$), 2.4 (s, 3H, $CH_3$)

**Solid product: $^1$H NMR** (400MHz, $D_2O$) δ ppm: 7.8 (d, 2H, *J* = 8.2, Ar-H), 7.3 (d, 2H, *J* = 8.1, Ar-H), 2.6 (s, 3H, $CH_3$), 2.3 (s, 3H, $CH_3$)

**Scheme 3.** Possible sources of synthesis error from side reactions. A shows a side reaction between sodium methoxide and water, while B shows the irreversible step in the Claisen Condensation.



Analyses of resulting products were all done by H-NMR at 400MHz, since no further analysis was needed to characterize the product.

Since there is no diketone in any of the reactants, the peak representing the two hydrogens on the carbon between the diketone are the most critical in the H-NMR characterization **(Scheme 1)**. None of the NMRs taken from the any of the 3 reactions contained the critical 5.96ppm peak. As a result, it was concluded that none of the reactions worked.

Possible confounding factors could vary widely. First, one of the most critical reagents in the reaction is the basic properties of the sodium methoxide reagent. A dry environment was incorporated in the second and third reaction mechanisms in order to protect the reaction from side reactions.[9] However, the sodium methoxide reactant was not tested individually. It is possible that it was contaminated, and therefore unable to act as a base in all three reactions. Second, further investigation into the "driving" deprotonation step of the Claisen Condensation could also be a source of error in the synthesis mechanism. In the current synthesis method, at the deprotonation step of the Claisen Condensation, there are 2 possible sources of base: leftover sodium methoxide with pKa ~ 16, and the ethyl oxide leaving group with pKa ~ 16.[7]

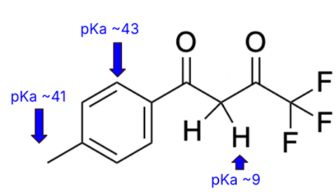



**Figure 3.** Graphic showing the pKa of all hydrogens on the wanted product: 4,4,4-trifluoro-1-(4-methyl-phenyl)-butane-1,3-dione.

From Figure 3, it is clear that a base with pKa larger than 9 but less than 41 is sufficient to deprotonate the carbon between the diketone and drive the reaction forward until it is irreversible.[10] As a result, adding an extra base that is strong such as $NH_2$ with pKa ~ 36 or NaH with pKa ~ 35 could help push this reaction forward.[7]

## AUTHOR INFORMATION

### Author Contributions

The manuscript was written through contributions of all authors. / All authors have given approval to the final version of the manuscript.

## ACKNOWLEDGMENT

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## SUPPORTING INFORMATION

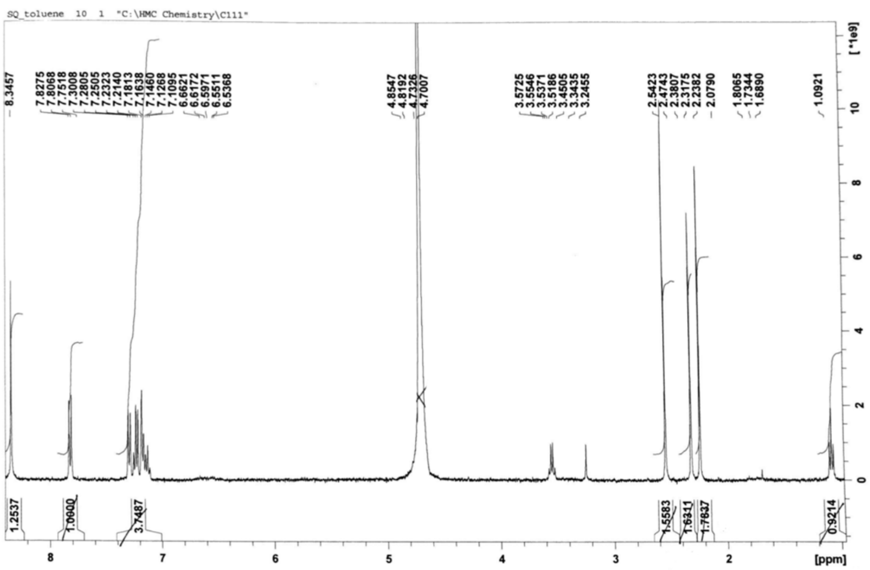
*Mechanism 1:* $^{1}$**H NMR** (400MHz, $D_2O$) δ ppm: 8.3 (s, 2H, Ar-H), 7.2 (d, 2H, *J* = 8.3, Ar-H), 6.5 (m, 7H, Ar-H), 2.5 (s, 3H, $CH_3$), 2.3 (s, 3H, $CH_3$), 2.1 (s, 3H, $CH_3$), 1.1 (t, 2H, *J* = 7.2, $CH_3$)

*Mechanism 2:* $^{1}$**H NMR** (400MHz, $D_2O$) δ ppm: 8.3 (s, 10H, Ar-H,$CH_3$), 7.8 (d, 2H, *J* = 8.1, Ar-H), 7.3 (d, 2H, *J* = 7.0, Ar-H), 2.6 (s, 3H, $CH_3$), 2.3 (s, 3H, $CH_3$)

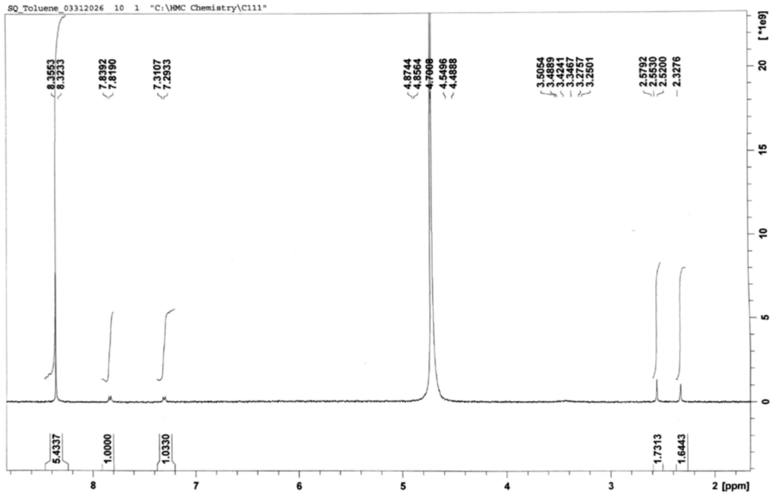
*Mechanism 3 Liquid:* $^{1}$**H NMR** (400MHz, $CDCl_3$) δ ppm: 7.9 (d, 2H, *J* = 8.2, Ar-H), 7.3 (d, 2H, *J* = 8.0, Ar-H), 2.6 (s, 3H, $CH_3$), 2.4 (s, 3H, $CH_3$)

*Mechanism 3 Solid:* $^{1}$**H NMR** (400MHz, $D_2O$) δ ppm: 7.8 (d, 2H, *J* = 8.2, Ar-H), 7.3 (d, 2H, *J* = 8.1, Ar-H), 2.6 (s, 3H, $CH_3$), 2.3 (s, 3H, $CH_3$)

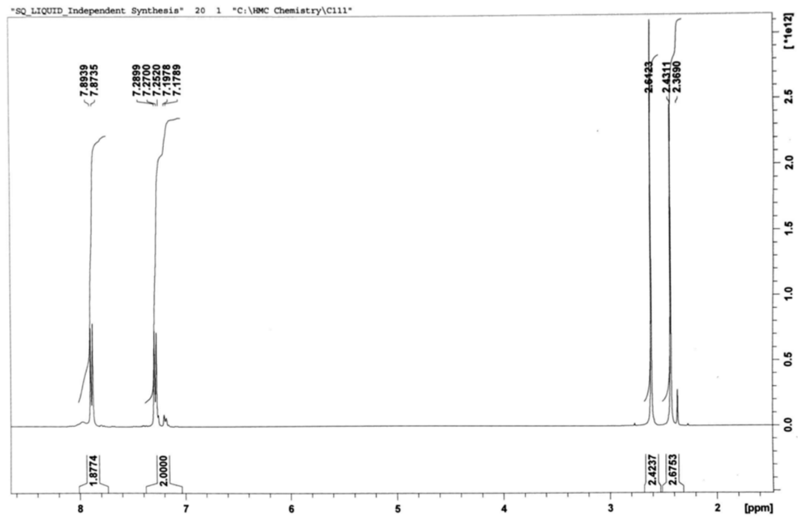
*Mechanism 1*




*Mechanism 2*




*Mechanism 3 Liquid*




*Mechanism 3 Solid*